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Digital entrepreneurial platform capability: Instrument development and validation, and its impact on entrepreneurial performance

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ABSTRACT

Digital platforms have increasingly become pivotal tools in the entrepreneurial landscape. A significant body of literature on digital entrepreneurship has conceptualized the role of digital platforms in enabling entrepreneurial activities. However, despite this conceptual foundation, empirical evidence on how digital platforms enhance entrepreneurial performance remains limited, and there is no adequate instrument for measuring digital platform capabilities in the entrepreneurial context. Therefore, this paper has a twofold purpose: first, to develop and validate a multidimensional scale for measuring digital entrepreneurial platform capabilities from the entrepreneurs' perspective; and second, to assess the impact of these platforms on entrepreneurial performance. Using a psychometric approach to scale development, this research develops and validates a 23-item, four-dimensional scale for measuring the capabilities of digital platforms. The nomological validity of the developed instrument is then evaluated by examining the direct link between digital entrepreneurial platform capabilities and entrepreneurial performance. These findings contribute to existing studies on digital entrepreneurship by developing a comprehensive measure of platform capabilities and offering entrepreneurs practical strategies for leveraging digital platforms to optimize performance in the digital entrepreneurship ecosystem.

1. Introduction

Digital entrepreneurship has emerged as an evolutionary notion of entrepreneurship, emphasizing the transformational role of digital technologies in entrepreneurial activities (Jones et al., 2023; Olan et al., 2024). Digital platforms are among the most significant technologies that revolutionize the entire process of entrepreneurial venture creation (Bachmann et al., 2024; Nambisan et al., 2019). Digital platforms take on a multi-faceted role in entrepreneurship that can serve as enablers, contexts, and outcomes (Chalmers et al., 2021; Nambisan, 2017; von Briel et al., 2018).

First, entrepreneurs can deploy various digital platforms to obtain and exploit necessary resources and solutions that enable them to start and grow their entrepreneurial ventures (Jones et al., 2023; Kyriakidis & Tsafarakis, 2024). Digital platforms are no longer just a way to expand existing processes and enhance accessibility, scalability of knowledge, and market reach, but they have also been substitution mechanisms (Rodríguez-Covarrubias & Álvarez-Figueroa, 2024; von Briel et al., 2018). For instance, platforms like crowdsourcing and crowdfunding have emerged as alternative online communities, replacing traditional entities such as incubators and conventional investors, thereby enabling entrepreneurs to effectively access different resources (Bellesia et al., 2019; Lall et al., 2022; Santos et al., 2024).

Second, digital entrepreneurial platforms have become new contexts or environments where there are entrepreneurial opportunities for emerging innovative business models (Y. Lin et al., 2025). Digital platforms have transformed the process of idea-to-venture by creating environments supporting idea sharing, opportunities identification and validation, problem-solving, fostering innovation, and venture creation (Ratten, 2023; S. C. Santos & Neumeyer, 2023). Third, the relevance of

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digital platforms in entrepreneurial activities is also reflected in expanding the scope and impact of entrepreneurial outcomes. Digital platforms enable entrepreneurs to expand the features, scope, and market presence of their product or service offerings (Etemad, 2023; Nambisan, 2017). By exploiting digital platforms' capabilities such as the co-creation of knowledge and ideas, and collective communities of diverse actors, entrepreneurs are more likely to adjust and innovate their offerings to be aligned with market demands (Rayna et al., 2015).

Despite the growing body of entrepreneurship research focused on conceptualizing digital entrepreneurship (e.g. Fernandes et al., 2022; Olan et al., 2024; Sahut et al., 2021), there is a paucity of research on measuring the capability of digital entrepreneurial platforms in entrepreneurial processes and activities. Although several studies have used different instruments to measure digital platforms capabilities, these instruments have measured the capabilities of entrepreneurs' platforms to access and exchange information and knowledge with their partners using two key capabilities, which are platform integration and reconfiguration (Cenamor et al., 2019; Jiang et al., 2023). However, the concept of a digital entrepreneurial platform is much more comprehensive, extending beyond just internal platforms developed within an enterprise to streamline its operations, to encompass open ecosystems that bring together multiple stakeholders across industries, enabling interactions and transactions on a broader scale (Hsieh & Wu, 2019; Rodrigo et al., 2025). Hence, there remains a notable lack of comprehensive and context-specific scales that can measure the capabilities of digital entrepreneurial platforms.

On the other hand, there is still a lack of empirical evidence regarding the role of digital entrepreneurial platforms in accomplishing entrepreneurial endeavors. While prior studies have placed significant emphasis on theorizing and conceptualizing how digital platforms facilitate entrepreneurial actions such as exploring ideas, fostering innovation, and accelerating market entry (Bellesia et al., 2019; Srinivasan & Venkatraman, 2018), quantitative studies remain limited. In addition, these studies have discussed entrepreneurial opportunities provided by platform-based innovation ecosystems, with examples and practical implications (Chandna, 2022; Hsieh & Wu, 2019a)). However, few scholarly studies have quantitatively explored the role of digital entrepreneurial platforms in driving entrepreneurial performance (Cenamor et al., 2019; Elia et al., 2020; Usman & Sun, 2023). As such, recent studies such as the study of Fernandes et al. (2022) argue that additional empirical and detailed evidence is needed to investigate how digital platforms can enhance entrepreneurial performance. As concluded by Gregori et al. (2024), further quantitative research on the role of digital entrepreneurial platforms in driving entrepreneurial performance could provide valuable insights into the impact of different digital entrepreneurial models on customer satisfaction and the overall success of ventures.

Therefore, this study has a twofold purpose; firstly, this study takes the initiative to develop measurement scales for digital entrepreneurial platforms capability. Based on the psychometric procedure of scale development, the study aims to systematically develop and validate a multidimensional scale to measure the utilization of digital entrepreneurial platforms from the entrepreneurs' perspective in the digital entrepreneurship ecosystem. Secondly, this study also aims to explore the role of digital entrepreneurial platforms in driving entrepreneurial performance. In other words, this study contributes to the literature by addressing the following key research questions.

RQ1. How can a comprehensive and context-specific scale be developed and validated to measure the capabilities of digital entrepreneurial platforms?

RQ2. What is the role of digital entrepreneurial platforms in enhancing entrepreneurial performance within the digital entrepreneurship ecosystem?

Furthermore, the findings of this study offer practical insights for entrepreneurs, platform developers, and policymakers on how they can leverage digital platforms to drive entrepreneurial processes and activities, thereby achieving entrepreneurial performance within the digital entrepreneurship ecosystem.

2. Theoretical background

2.1. Digital entrepreneurship

Digital technologies have emerged as crucial factors that not only impact the essence of innovation but also redefine the fundamental aspects of entrepreneurship and how it is conducted (Polyakov & Kovshun, 2021). Digital entrepreneurship constitutes a transformative wave of entrepreneurial activity empowered by digital technologies. Digital entrepreneurship, as an emerging concept, is extensively used in literature to reflect the act of starting up, managing, and scaling up entrepreneurial ventures by leveraging the new opportunities provided in the digital space (Davidson & Vaast, 2010, pp. 1–10; E. A. dos Santos & Torkomian, 2021). The hallmark of digital entrepreneurs is their extensive use of state-of-the-art digital technologies to explore innovative ways to conduct their business, develop innovative business models, and refine business intelligence (Sahut et al., 2021). The digitalization of entrepreneurial processes has facilitated the removal of boundaries between different phases of entrepreneurship, resulting in a notable decrease in barriers from invention to innovation (Nambisan, 2017; Sahut et al., 2021; Steininger, 2018). Consequently, the focus of recent literature on DE has extensively been on how entrepreneurs can effectively utilize digital technologies to scale their ideas into successful ventures (Neumeyer et al., 2021; Troxler & Wolf, 2017).

Within the realm of entrepreneurship, digital technologies have been placed into three interrelated elements, namely, digital artifacts, digital platforms, and digital infrastructures, providing ample entrepreneurial opportunities (Nambisan, 2017). A digital artifact, as described by Nambisan (2017), refers to any digital element, such as software, applications, or media content, that is integrated into a new product or service to deliver a distinct function or benefit to the end-user. Virtual reality is a notable example of a digital artifact that can present entrepreneurial opportunities for various industries such as healthcare, aviation, tourism, or sports. Secondly, a digital platform is characterized as a collective and standardized framework of services and architecture designed to facilitate the hosting of complementary offerings, encompassing digital artifacts. For example, Kickstarter is one of the crowdsourcing platforms that help entrepreneurs not only to secure financial support but also to bring and validate their innovative ideas. Thirdly, Digital infrastructure comprises a range of technological tools and systems that offer computing, collaboration, and/or communication capabilities to facilitate and bolster innovation and entrepreneurship such as social media, data analytics, cloud computing, online communities, and 3D printing (Braune & Dana, 2022).

2.2. Digital entrepreneurial platforms

Platformization as a technological emergent trend has completely transformed business models, especially for entrepreneurs (Nambisan et al., 2018a; Troxler & Wolf, 2017). Digital platforms have become a generative mechanism for business opportunities, value-creation processes, and entrepreneurial activities (Troxler & Wolf, 2017). So recently, much research on entrepreneurship has focused largely on the vital question that how entrepreneurial pursuits are facilitated by digital platforms (Nambisan et al., 2018). Online entrepreneurial platforms refer to digital environments that are composed of a diverse range of networks involving varied actors, resources, and activities to facilitate user-driven innovation processes (Baran & Berkowicz, 2021). Based on a socio-technical approach, digital entrepreneurial platforms can be conceptualized as a digital ecosystem that facilitates the transactions and the interaction of entrepreneurs, users, and other stakeholders in

such a way that promotes effective entrepreneurial activity (Song, 2019). Digital platforms bring together technical tools and social connections to support the entrepreneurial process. Therefore, a digital entrepreneurial platform is a construct that can be used to reflect the degree of usage of digital platforms in supporting the "idea-to-venture process."

Digital platforms have dynamic potential and mechanisms reshaping the entrepreneurial landscape and fueling a wave of innovation and growth. The success of digital platforms relies on value-creation mechanisms that play a crucial role in facilitating seamless transactions and providing the necessary tools and solutions to foster innovation within the platform ecosystem (Baran & Berkowicz, 2021). From an economic perspective, digital platforms are seen as "two-sided markets" or "multi-sided markets" that act as intermediaries and facilitate exchange between different types of users who would not be able to interact with each other otherwise (Gawer, 2014). In this context, the success and attractiveness of digital platforms for entrepreneurs depends on the idea which is defined as "network effects" that manifest when the usage of a digital platform increases as its user base expands (Thies et al., 2018). With the presence of strong network effects in a digital platform, entrepreneurs can expand their reach and deliver their offerings to a vast customer base (Srinivasan & Venkatraman, 2018). Additionally, the power of digital platforms increasingly relies on the notion of "collective intelligence" which serves as a fundamental pillar for these platforms (Elia et al., 2020). The collective intelligence of digital platforms nurtures a supportive ecosystem for entrepreneurs, bringing together diverse groups of individuals with varying backgrounds, expertise, and interests to exchange solutions, knowledge, and ideas (Liu et al., 2018; Palacios et al., 2016). For instance, crowdsourcing platforms like InnoCentive, now known as the Wazoku Crowd, offer open and collaborative communities of experts and innovators (Acar, 2019). These platforms generate and present innovative solutions and ideas to address the challenges that entrepreneurs encounter throughout their journey from idea to venture (Foege et al., 2019; Palacios et al., 2016).

Although platforms share certain fundamental dynamics, each platform has different ways of being applied and organized in the market. As a result, Evans and Gawer (2016, pp. 1–30) suggested categorizing digital platforms based on their principal purpose into four distinct types, including, transaction platforms, innovation platforms, integrated platforms, and investment platforms. First, the main functionality of transaction platforms is to facilitate social or commercial transactions between various users (Mady et al., 2022). These platforms provide a marketplace or digital space where products, services, or information can be exchanged. Uber, Airbnb, eBay, and Amazon serve as prime examples of successful transaction platforms. Second, innovation platforms represent a digital ecosystem that enables developers, innovators, and entrepreneurs to collaborate, develop, validate, and refine their innovative ideas, products, and services. There are numerous digital platforms that play a crucial role in supporting and fostering innovation across various industries, particularly for nascent entrepreneurs such as Kickstarter, Indiegogo, and Wazoku Crowd. Third, integrated platforms are those platforms that seamlessly merge the essential functionalities of transactional and innovation platforms. Salesforce, Amazon, and Shopify are a case in point, serving as online marketplaces, coupled with providing solutions that enable entrepreneurs to build, launch, and grow their businesses. Fourth, the focus of investment platforms has been on facilitating investment-related activities. Crowdfunding platforms are conceived of as modern digital alternatives to traditional financial sources such as bank loans.

Prior literature, as shown in Table 1, on digital entrepreneurial platforms has addressed various capacities, including opportunity identification, resource mobilization, fostering innovation, and customer engagement. Nevertheless, a growing body of research has either focused on theoretical frameworks or adapted scales originally designed for broader information technology capabilities, as demonstrated by Cenamor et al. (2019) and Jiang et al. (2023). These scales do not fully measure the unique functionalities and dynamic nature of entrepreneurial digital platforms. Moreover, empirical validations of what these capabilities are and how they assist entrepreneurs in diverse entrepreneurial contexts remain limited. To address these gaps, the current study develops and validates a dedicated scale for entrepreneurial digital platform capabilities.

2.3. Digital entrepreneurial platform capability as a multidimensional construct and its linkage with entrepreneurial performance

Digital entrepreneurial platforms are conceptualized in this study as digital environments that help entrepreneurs facilitate various entrepreneurial functionalities, including the creation of new opportunities, sharing resources, fostering innovation, and supporting effective marketing strategies. Therefore, the construct of 'Digital entrepreneurial platform capability' can be conceptualized as a multidimensional construct, indicating the inherent capacity or potential of digital entrepreneurial platforms to help entrepreneurs achieve their entrepreneurial pursuits. Through reviewing the literature, it is evident that digital platforms can enable entrepreneurs to perform four key functionalities.

2.3.1. Opportunity identification & validation

Digital platforms (e.g., crowdsourcing and crowdfunding platforms) consist of a spatially dispersed crowd that can actively contribute to the co-creation of opportunities for new and innovative products, services, and ideas with global potential (Ahsan & Musteen, 2021; Ojala et al., 2018). Entrepreneurial opportunity is acknowledged as a pivotal concept in the field of entrepreneurship, referring to potential situations that can be transformed into successful ventures (Ardichvili et al., 2003; Kirzner, 1997). For emergent entrepreneurial opportunities to be accepted and successful in the market, they must possess unique and unconventional characteristics. As per classic entrepreneurship theory (Kirzner, 1973), the temporal and spatial divergence of knowledge fosters entrepreneurial opportunities. This perspective emphasizes that the expansion and dispersion of knowledge over time and space play an essential role in identifying and exploiting entrepreneurial opportunities (Ojala et al., 2018). In the digital era, participation on digital platforms with a spatially dispersed crowd (such as Amazon Mechanical Turk) contributes to expanding an entrepreneur's collective and diverse knowledge, facilitating the opportunity creation process (Dellermann et al., 2020). Many prior studies accentuated that digital platforms facilitate the opportunity identification and effective adaption to fast-changing market needs (Cenamor et al., 2019; Swartz et al., 2022). Palacios et al. (2016) and Acar (2019) indicate that entrepreneurs are increasingly harnessing the power of the crowd for idea generation, discovering innovative solutions, and securing funding for new ventures to sustain their entrepreneurial endeavors.

The collective intelligence approach is conceived of as a concrete solution for nascent entrepreneurs to deal with uncertainty intertwined with the opportunity creation process (Dellermann et al., 2020; Maine et al., 2015). Dellermann et al. (2020) suggested that most entrepreneurial opportunities are more likely to be created than recognized, as they rely mostly on the knowledge, actions, and imagination of an entrepreneur. As per opportunity creation perspective, the decision-making context in which these opportunities arise is described as highly uncertain (Maine et al., 2015). To deal with the uncertainty inherent in the opportunity-creation process, entrepreneurs must enact and validate their initial opportunities and ideas while interacting with knowledgeable others in a social setting such as receiving feedback from consultants or peers (Nambisan et al., 2018a; Tocher et al., 2015). Therefore, the collective intelligence of digital platforms can help entrepreneurs provide diverse, scalable, and resilient networks beyond their immediate contacts, enabling socially constructed co-creation for entrepreneurial opportunities.

Table 1
Literature review synthesis.

Study (Author (s), Year)	Focus/Objective	Type of Study	Sample/Data	Digital Capabilities Addressed	Findings/Key Insights	Research Gap
Srinivasan and Venkatraman (2018)	Theorize how digital entrepreneurs leverage resource networks to access financial capital and human talent for success on digital platforms.	Theoretical study	Conceptual framework based on prior literature	- Digital Resource Mobilization - Network-Oriented Digital Platforms	Highlight the criticality of high-status venture capital relationships to access essential resources for entrepreneurial success.	Lack empirical validation for proposed resource network strategies.
Nambisan et al. (2018)	Examine how open innovation and platform ecosystems transform entrepreneurship, emphasizing co-creation and resource integration.	Conceptual essay	Literature synthesis on open innovation and digital platforms	- Platforms for Collaborative Innovation - Ecosystem-Driven Resource Integration	Propose that digital platforms facilitate collaborative innovation and resource-sharing networks	Focus on theoretical frameworks; lack empirical validation of proposed mechanisms for fostering innovation and resource mobilization.
Thies et al. (2018)	Investigate the role of network effects and input controls on the success of crowdfunding platforms.	Exploratory empirical study	Time series data from 339,721 Kickstarter campaigns (2009–2017)	- Sharing and accessing financial resources for entrepreneurs	Explore how crowdfunding platforms enable entrepreneurs to access financial support and how network effects influence platform growth and dynamics.	Lack exploration of fostering innovation and comprehensive customer engagement strategies beyond crowdfunding contexts.
Hsieh and Wu. (2019)	Propose a framework categorizing platform-based innovations for entrepreneurs in the invention and commercialization stages.	Conceptual study	Framework synthesized from platform and innovation literature	- Facilitating Platform-Based Innovation - Supporting Digital Commercialization Processes	Highlight how platforms enable entrepreneurial innovation by fostering modularity and recombination in product and service creation.	Lack empirical testing of the framework and broader exploration of marketing/customer engagement.
Bellesia et al. (2019)	Explore how gig workers construct professional identities through interactions on online labor platforms.	Qualitative research	46 interviews with freelancers on a popular online labor platform	- Facilitating Professional Identity Formation - Enabling Digital Workforce Engagement	Online labor platforms enable workers to create professional identities through platform-mediated interactions and self-presentation.	Limited to gig workers and online labor platforms; does not address broader entrepreneurial capabilities like resource mobilization or fostering innovation.
Cenamor et al. (2019)	Investigate how entrepreneurial SMEs enhance performance through digital platforms, focusing on digital platforms and network capabilities.	Quantitative research	Survey, 230 Swedish SMEs	Platform integration; Platform reconfiguration	Demonstrate that digital platform capabilities enhance network capabilities, which in turn improve SMEs' performance. Exploitation and exploration logics moderate these effects.	The study uses a scale for IT capabilities adapted to measure digital platform capabilities, which may not fully capture the unique functionalities of digital entrepreneurial platforms.
Dellermann et al. (2020)	Explore how collective intelligence and crowdsourcing can enhance opportunity creation by reducing cognitive and resource constraints for entrepreneurs.	Conceptual study	Conceptual arguments and propositions based on existing literature	- Opportunity Identification & Validation - Resource Mobilization & Sharing	Demonstrate how crowdsourcing provides access to diverse, heterogeneous knowledge and flexible resources, aiding in iterative development, stakeholder persuasion, and resource mobilization.	Limited exploration of empirical validation; focus is primarily on theoretical arguments without data-driven analysis. Other capabilities like fostering innovation and customer engagement are underexplored in this context.
Fan et al. (2021)	Investigate the time-contingent learning processes of habitual entrepreneurs in digital platform ecosystems.	Empirical study	Facebook app development data from 2007 to 2008	- Opportunity Identification - Resource Mobilization	Identify how dynamic platform ecosystems create unique challenges for entrepreneurial learning.	Limited exploration of other digital capabilities, such as fostering innovation and customer engagement. The study is focused narrowly on learning effects in Facebook app development.
Baran and Berkowicz (2021)	Propose a conceptual model integrating living labs with digital platform ecosystems for sustainable entrepreneurship.	Conceptual/ Theoretical study	Theoretical synthesis of prior literature.	- Fostering Innovation - Opportunity Identification and Validation (partial)	Introduce living labs to co-create solutions for sustainable innovations.	The model is theoretical and requires empirical validation. Does not focus explicitly on marketing or customer engagement.
Bonina et al. (2021)	Explore the foundational concepts and developmental implications of digital platforms, differentiating between transaction and innovation platforms.	Conceptual study and literature review	Literature from ICT and information systems research (49 articles)	- Transaction Platforms for Developmental Interactions - Innovation Platforms for Indigenous Service Development	Define transaction and innovation platforms and their roles in socio-economic development, focusing on their socio-technical nature and ecosystem dynamics.	Lacks empirical validation of platform typologies. Calls for further exploration of indigenous innovation and platform governance
Swartz et al. (2022)	Examine the determinants of opportunity development among female founders of digital platform start-ups in South Africa.	Qualitative research	In-depth interviews with women founders of digital platform start-ups	Entrepreneurial opportunity identification and development	Women entrepreneurs can use education, corporate experience, and digital platforms to identify opportunities, driven by life transitions and a desire for social change.	Existing models overlook the role of multiple capabilities in digital platform entrepreneurship.

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Table 1 (*continued*)

Study (Author (s), Year)	Focus/Objective	Type of Study	Sample/Data	Digital Capabilities Addressed	Findings/Key Insights	Research Gap
Bhatti et al. (2022)	Examine the role of digital platform capabilities in enhancing supply chain innovation and firm performance in MSMEs.	Quantitative, Structural Equation Modeling (SEM)	221 MSMEs in the manufacturing sector	- Integration capabilities for knowledge exchange - Reconfiguration capabilities	Digital platform capabilities play a crucial role in improving financial performance through supply chain innovation.	Measure only two main digital platform capabilities but overlook other important capabilities for entrepreneurs, such as customer engagement and innovation fostering.
Lall et al. (2022)	Investigate the role of online mentoring platforms in supporting entrepreneurs through a field experiment.	Field experiment	Entrepreneurs and mentors on an international online mentoring platform	- Resource Mobilization and Sharing (partial) - Marketing and Customer Engagement (partial	Demonstrate the importance of mentorship for resource mobilization and opportunity recognition. Highlights the barriers entrepreneurs face in engaging with mentors on digital platforms.	Limited to mentorship and user engagement; lacks broader exploration of innovation or systematic marketing mechanisms within digital platforms.
Fernandes et al. (2022)	Systematically review literature on digital entrepreneurship to identify key themes, and gaps, and propose a holistic research agenda.	Systematic literature review	83 articles from the Web of Science database	- Digital Opportunity Identification and Success Factors - Ecosystem-Driven Resource Mobilization - Co-creation on Digital Platforms	Identify five thematic approaches: success factors, ecosystems and smart cities, business models, sharing platforms, and co-creation. Proposes a holistic model for digital entrepreneurship research.	The study lacks empirical validation of the proposed framework.
Ratten (2023)	Investigate how the ClickforVic digital platform facilitated transformational entrepreneurship among farm entrepreneurs during the COVID-19 crisis.	Qualitative study	Semi-structured in-depth interviews with farm entrepreneurs	- Facilitating new relationships - Information Acquisition - Connect with customers	Digital platforms enabled farm entrepreneurs to achieve financial, social, and community goals during the crisis.	Limited to a single digital platform and region. Calls for broader studies on the role of digital platforms across varied contexts.
Jiang et al. (2023)	Examine the impact of SMEs' digital platform capabilities (DPC) on their innovation performance, moderated by ecological institutional norms (EIN).	quantitative study	Survey data from 346 Chinese SMEs	Platform integration; Platform reconfiguration	Digital platform capability enhances innovation through resource integration and value co-creation.	The study uses a scale for IT capabilities adapted to measure digital platform capabilities. The study also focuses on digital platforms owned by SMEs, neglecting the capabilities of external or shared platforms that entrepreneurs might leverage.
Gregori et al. (2024)	Investigate how sustainable entrepreneurs enact digital connectivity on platforms to create and capture value, integrating socio-materiality and business models.	Qualitative study	Visual and textual data from 19 sustainable entrepreneurial initiatives on YouTube	- Digital Connectivity for Value Creation - Platform-Driven Socioenvironmental Co-Creation - Digital Mechanisms for Multidimensional Value Capture	Highlight how digital platforms enable connectivity for socioenvironmental change. Connectivity is co-created by human and nonhuman actors (e.g., algorithms, content). Entrepreneurs leverage connectivity for financial and non-economic value capture while adapting to platform	Further research is needed to explore platform capabilities for broader entrepreneurial contexts.

2.3.2. Resource mobilization & sharing (knowledge/capital)

Entrepreneurial start-ups, similar to small and medium enterprises (SMEs), face a range of challenges, including limited resources, technology-related difficulties, and accessing cutting-edge knowledge (Hsieh & Wu, 2019). Based on the social network perspective, extensive research on entrepreneurship offers a valuable understanding of how strong social networks contribute to businesses by providing a wide range of unique and non-overlapping information and resources (Chen & Tan, 2009; Slotte-Kock & Coviello, 2010). As Hite (2005) indicated that social networks play a vital role in entrepreneurship by serving as pathways for entrepreneurial firms to access resources and opportunities. Using digital platforms leads to a redefinition of the nature of the networks to which entrepreneurs are affiliated (Nambisan et al., 2018a). Besides social and professional networks offered on digital platforms like Facebook and LinkedIn, entrepreneurs can also be members of other digital networks and communities(Srinivasan & Venkatraman, 2018). These platforms provide network capability with enormous potential to access diverse resources, including, knowledge, business tools and technologies, and funding and investment opportunities (Cenamor et al., 2019; Liu et al., 2018).

Digital platforms are instrumental in facilitating knowledge sharing in various domains. These platforms enable entrepreneurs to connect with diverse communities of fellow entrepreneurs, experts, innovators, incubators, and mentors, providing valuable guidance and in-depth knowledge (Hsieh & Wu, 2019). Knowledge resources can be valuable solutions that contribute to solving problems encountered by entrepreneurs (Bhatti et al., 2022; Malhotra & Majchrzak, 2019; Palacios et al., 2016). Knowledge resources can also encompass information about market trends, consumer preferences, customer feedback, and industry insights that drive entrepreneurial pursuits (Liu et al., 2018). Additionally, digital platforms that are based on the idea of collective intelligence, such as WhataVenture, have the potential to develop skills and personal capabilities, enabling entrepreneurs to make informed decisions and drive the growth of their ventures (Cenamor et al., 2019).

Over the recent decade, the alternative finance system has seen rapidly evolving in providing a plethora of innovative financial instruments and services thanks to digital transformation. Crowdfunding platforms have arisen as an alternative financial solution for funding ideas and innovations (Rey-Martí et al., 2019; Thies et al., 2018). A crowdfunding platform is a method of financing that combines the principles of microfinance and crowdsourcing (Yasar, 2021). Microfinance reflects those small loans and financial services provided to those who face difficulty in having access to traditional financial services, whereas crowdsourcing refers to sharing and collecting different aspects of contributions (services, content, or ideas) from a large community of people, usually through online platforms. Crowdfunding, therefore, is a form of alternative financing that allows small investors to pool their resources and contribute to a project or idea in exchange for a potential financial return or other rewards. Crowdfunding platforms stand out as a powerful way to fund entrepreneurial ideas and ventures ignored by traditional sources of capital (Cicchiello, 2019; Thies et al., 2018). In addition, these platforms adopted a risk and profit-sharing approach to enable entrepreneurs to share their venture risks, which are characterized by high, among a large number of funders (Cicchiello, 2019; Nambisan et al., 2018).

2.3.3. Fostering innovation

Literature on entrepreneurship accentuates the interrelationship of innovation and entrepreneurship. Innovation is a paramount aspect of the success of the entrepreneurial process, reflecting the process of transforming entrepreneurial ideas or inventions into new products and services which is the market value for both customer and entrepreneur. In addition, an existing set of entrepreneurial skills is also crucial to the success of the innovation process. Thus, entrepreneurs are often recognized for their ability to think creatively, identify new opportunities, and transform innovative ideas into successful business ventures. Digital platforms can serve as important catalysts and facilitators of innovation and entrepreneurship for those entrepreneurs who are looking to develop and launch new ideas and ventures (Brem et al., 2019; Yang & Han, 2021). The power of digital platforms to empower the innovation process lies in their ability to create and share heterogeneous knowledge extracted from diverse sources (Ben Arfi & Hikkerova, 2019). This potential enables these platforms to facilitate the exchange of valuable insights, fostering innovation and driving forward progress (Cenamor et al., 2019; Wareham et al., 2014)

Digital platforms can provide entrepreneurs with an environment fostering their innovation activities in various ways. First, digital platforms, particularly innovation platforms rooted in the concept of collective intelligence, offer a valuable resource for generating ideas (Nambisan et al., 2019; Yang & Han, 2021). By facilitating connections and fostering the sharing of experiences among experts and entrepreneurs from diverse backgrounds, these platforms serve as a catalyst for idea generation and improvement (Palacios et al., 2016. Second, these platforms also play a significant role in innovating new products and services, empowering consumers to actively contribute as co-producers and co-designers of their experiences through online customer participation (Hsieh & Wu, 2019; Srinivasan & Venkatraman, 2018). Third, Entrepreneurs can access a wide range of general and comprehensive services through certain platforms (Braune & Dana, 2022). Take Amazon Web Services (AWS) as an example, which offers start-up enterprises an array of services that encompass cloud computing, data management, cost management, and business productivity. These services can be effective solutions for entrepreneurs' innovation needs.

2.3.4. Marketing & customer engagement

Commercializing new products and services is one of the most critical functions of digital platforms (Hsieh & Wu, 2019). Digital platforms, viewed through an economic lens, are considered intermediaries in "two-sided markets" or "multi-sided markets". These platforms enable the exchange between various user groups who may not have been accessible otherwise (Gawer, 2014). While entrepreneurs may have the potential to create novel products or services, they often encounter challenges when it comes to effectively commercializing them (Hsieh & Wu, 2019). Digital platforms play a crucial role in helping entrepreneurs bring their brands to the digital marketplace such as Facebook Marketplace, and Amazon Marketplace. These platforms enable them to expand their customer base, increase brand visibility, and ultimately grow their business in the digital realm (Nambisan et al., 2018a). Digital platforms enable entrepreneurs to access market trends and respond quickly to market needs (Ahsan & Musteen, 2021; Cenamor et al., 2019). These platforms provide entrepreneurs with valuable insights into consumer behavior, market demand, and emerging trends (Mady et al., 2022). Entrepreneurs can utilize the data and analytics provided by digital platforms to make knowledgeable choices regarding their products, services, and marketing strategies (Sivarajah et al., 2020). They can identify market gaps, understand customer preferences, and adapt their offerings accordingly. Moreover, digital platforms enable entrepreneurs to reach a wider audience and respond rapidly to changing market conditions (Etemad, 2023).

Considering that entrepreneurial digital platforms (DEPs) serve as a digital environment for an enterprise and having access to these DEPs results in enhanced entrepreneurial performance since these DEPs empower entrepreneurs by enabling them to identify and validate newer potential business ideas while fostering innovation within the enterprise through resource mobilization and knowledge sharing. Further, access to DEPs also helps enterprises connect with potential customers and leverage this customer network to gain a wider knowledge of the market trends. Simply put, being able to squeeze newer business opportunities while fostering innovation, and optimal resource mobilization results in increased productivity through streamlined operations which is further leveraged by wider knowledge of the market trends through increased connectivity with the customers helping the enterprise increase market

share; consequently, leading to an overall improved entrepreneurial performance. Thus, we hypothesize that.

H1. Access to DEPs significantly enhances entrepreneurial performance.

3. Methods and results

The study follows the psychometric procedure recommended for scale development in the fields of business, management, and marketing (Churchill, 1979; J. S. C. Lin & Hsieh, 2011). A five-stage process was employed in the development and validation of the measurement scales for the digital entrepreneurial platform (DEP) and entrepreneurial performance (EP), including item generation, purification of measurement scale, dimensionality reduction, validation of the scale, and nomological validation.

3.1. Stage 1: Item generation

A pool of potential items that reflect the construct of the digital entrepreneurial platform was initially generated through a two-step process; reviewing existing research literature and conducting in-depth interviews. The literature review serves two important functions: conceptualizing the construct and identifying the domains reflected in the construct (Churchill, 1979). Based on reviewing the literature addressing using digital platforms in the realm of entrepreneurship, the construct of digital entrepreneurial platform capability was conceptualized as a multidimensional construct, indicating the capabilities that digital entrepreneurial platforms can enable entrepreneurs to achieve their entrepreneurial pursuits across the idea-to-venture journey. By reviewing relevant literature, four dimensions have been identified to reflect the construct, which is entrepreneurial opportunities identification, resource mobilization and sharing (knowledge/capital), fostering innovation, and marketing and customer engagement. Table 2 illustrates an initial list of items generated by reviewing relevant literature and conducting interviews.

For entrepreneurial performance, this study aimed to use a measurement scale that could gauge the performance of an enterprise in terms of growth/improvement in its sales, profit, market share, cost-efficiency, and increase in its product/services portfolio over the past three years. Henceforth, the study operationally defines entrepreneurial performance as "A business's capability to achieve consistent growth in its sales volumes, and profitability, and expand its market share while maintaining the cost-efficiency and ensuring the increase in the number of products/services in the portfolio." Upon an extensive review of the literature, we were unable to find a measurement scale that could gauge entrepreneurial performance in the above-defined manner. Therefore, we borrowed items from different published studies (Ashraf et al., 2023; Guo et al., 2023; Maziriri et al., 2022; Zeb & Ihsan, 2020) and contextualized them in terms of language and subjectivity (subject to the above-mentioned business performance aspects). The list of the items is as follows (See Table 3).

3.2. Stage 2: Scale purification through expert review

Six interviews with entrepreneurs were conducted to further explore and validate the dimensions of DEP identified from the literature review. These interviews aimed to gather insights from entrepreneurs who have experience in leveraging digital platforms for their entrepreneurial activities. The interviews likely provided valuable insights and perspectives and real-life examples that complemented the findings from the literature review.

An expert panel review consisting of seven entrepreneurship scholars and three entrepreneurs was conducted to assess the existing items further to ensure face and content validity. Three criteria were used to evaluate the face and content validity of the scale items, including.

Table 2
Generation of measurement items for DEP.

Dimension name and items for DEP	Sources
Opportunity identification & validation	
1. Digital platforms allow me to recognize and seize new business opportunities I wouldn't have encountered otherwise.	Nambisan et al. (2018)
2. I rely on digital platforms to validate my early-stage entrepreneurial ideas.	Swartz et al. (2022)
3. The tools on these platforms offer effective mechanisms for scanning and assessing business opportunities.	Cenamor et al. (2019)
4. Thanks to digital platforms, I now have the opportunity to expand my business	Interviews
5. Digital platforms have opened up a world of opportunities for me	Interviews
6. Digital platforms truly transformed my perspective, leading me to embark on my own entrepreneurial venture.	Interviews
Resource mobilization & sharing	
1- Digital platforms are my primary source for accessing crucial entrepreneurial resources.	Cenamor et al. (2019)
2- The platforms offer technical capabilities that are instrumental in developing and refining my business offerings.	Cenamor et al. (2019)
3- Digital platforms have allowed me to expand my reach to supportive community of experts or entrepreneurs who can provide feedback and advice.	Hsieh and Wu (2019)
4- Analytics and performance metrics from these platforms are vital for monitoring and steering my business's success.	Sivarajah et al. (2020)
5- Digital platforms provide valuable knowledge and information.	Hsieh and Wu (2019)
6- Some digital platforms represent powerful source of capital	Thies et al. (2018); Cicchiello (2019)
7- Digital platforms help me collaborate closely and share resources, including software tools and technologies.	Interviews
8- Digital platforms help me to educate myself and stay updated on the latest trends and advancements in the digital space	Interviews
Fostering innovation	
1- Digital platforms are essential in facilitating the creation of innovative products and services in my venture.	Hsieh and Wu (2019)
2- I turn to digital platforms for innovative solutions when faced with entrepreneurial challenges.	Nambisan et al. (2019); Hsieh and Wu (2019)
3- Digital platforms inspire me to establish innovative solutions that operate entirely	Interviews
4- Digital platforms have been instrumental in helping me and my business identify challenges and create new innovations.	Interviews
Marketing & customer engagement	
1. Digital platforms are pivotal in my marketing efforts, allowing me to connect with and engage potential customers.	Etemad (2023) Gawer (2014)
2. The platforms provide the tools and channels necessary for effective product commercialization.	Etemad (2023)
3. I'm confident in utilizing digital platforms to expand my customer base and introduce my business ideas to the market.	Interviews
4. Digital platforms enable me to establish an online presence, showcase my services, and connect with clients effectively.	Interviews

1. Relevance: Does the item accurately capture the dimensions of digital entrepreneurial platforms' capability?
2. Clarity: Is the item clear and understandable?
3. Appropriateness: Is the item suitable for measuring digital entrepreneurial platforms across different types of platforms?

In this phase, each item was scored on a five-point Likert-type scale, spanning from "irrelevant", to "relevant", from "not clear" to "clear", and from "inappropriate" to "appropriate." Out of 22 items, 8 items were modified to enhance the clarity and suitability of the instrument. Only

Table 3
Scale items for entrepreneurial performance.

EP1: Over the three years, my business has witnessed a significant year-on-year growth in profit.
EP3: Over the three years, my business has witnessed a significant year-on-year growth in sales.
EP2: Over the three years, my business has achieved a higher market share compared to its peers.
EP4: Over the three years, my business has witnessed a significant increase in the number of products/services being offered.
EP5: Over the three years, my business has witnessed a significant improvement in the cost efficiency of production activities.

one item was added, and no new items were removed during this process. Then, 23 items were checked by 11 graduate students to avoid the problems related to wording and designing the questionnaire. Two items were subjected to modification, and no new items were added. Hence, 23 measurement items were created and subsequently chosen as the research instrument for stage 2 (see Table 4).

3.3. Stage 3: Dimensionality reduction through factor analysis (EFA)

Essentially, we conducted the exploratory factor analysis (EFA) for data dimensionality reduction. EFA assumes that a set of observable and measurable variables may be condensed into a few unobservable latent variables that share the variance and consequently, achieve dimensionality reduction(Bartholomew et al., 2011). More importantly, EFA condenses data in a way that helps in understanding the relationships and patterns identified through the observed shared variation (Fabrigar et al., 1999). The principal requirements for EFA include, among others, the necessity of having at least three variables for each factor to be considered (Williams et al., 2010), that Pearson's correlation coefficient r should be greater than 0.3, which implies that the correlation between variables is sufficiently strong (Williams et al., 2010), and a sufficient sample size. Concerning sample size there is controversy in the literature (Everitt, 1975; Fabrigar & Wegener, 2011). Hair et al. (2019) think observations above 100 is the sample size; whereas de Winter et al. (2009) opine that even if observations go below 50, then EFA can be conducted. Guadagnoli and Velicer (1988) opine that 150 observations will suffice and the dataset will produce factor loadings of more than 0.60. Comrey and Lee (1992) argue that every variable involved in factor analysis should be given 5–10 observations, while other authors conclude that arguments pertaining to a large number of observations are lacking in both theoretical as well as empirical basis (Fabrigar & Wegener, 2011). In an apt study (MacCallum et al., 1999), constructed a theoretical model that considered how sample size affects the recovery of factors, thereby providing some evidence for the claim that no such absolute thresholds exist for minimum sample size.

Table 4
Finalization of items after expert review.

Dimension name and items for DEP	After expert review
Opportunity identification & validation	
1. Digital platforms enable me to identify and capitalize on new business opportunities that may not have been accessible otherwise.	Modified
2. I rely on digital platforms to validate my initial business ideas.	Modified
3. Digital platforms provide tools that are effective for scanning and assessing business opportunities.	Modified
4. Digital platforms give me the opportunity to expand my business.	Modified
5. Digital platforms have opened up entrepreneurial opportunities for me.	Modified
6. Digital platforms have inspired me to pursue my own entrepreneurial endeavor.	Modified
Resource mobilization & sharing	
1. Digital platforms enable me to access crucial resources	Modified
2. Digital platforms provide me technical capabilities needed to develop and refine my business offerings.	Modified
3. Digital platforms have allowed me to expand my reach to supportive community of experts or entrepreneurs who can provide feedback and advice.	
4. Analytics and performance metrics from these platforms are vital for monitoring and steering my business's success.	
5. Digital platforms provide valuable knowledge and information.	
6. Some digital platforms represent powerful source of capital	
7. Digital platforms help me collaborate closely and share resources.	Modified
8. Digital platforms help me to educate myself and stay updated on the latest trends and advancements in the digital space	
Fostering innovation	
1. Digital platforms are essential in facilitating the creation of innovative products and services in my venture.	
2. I turn to digital platforms for innovative solutions when faced with entrepreneurial challenges.	
3. Digital platforms inspire me to establish innovative solutions that operate entirely	
4. Digital platforms have been instrumental in helping me and my business identify challenges and create new innovations.	
Marketing & customer engagement	
1. Digital platforms play a crucial role in my marketing strategies, enabling me to effectively connect with and engage potential customers.	Modified
2. The platforms provide the tools and channels necessary for effective product commercialization.	
3. I'm confident in utilizing digital platforms to expand my customer base.	
4. I'm confident in utilizing digital platforms to introduce my business ideas to the market.	Added
5. Digital platforms enable me to establish an online presence, showcase my services, and connect with customers effectively.	

Considering the above, the current study chose to have a sample size of 155 responses which is in adherence with 5:1 respondents-items ratio criteria. Using purposive sampling, we identified 211 Egyptian MSME firms situated in Cairo and Giza governorates (diversified across the businesses in agriculture, food and beverages, retail, health services, real estate, knowledge and ICT, education, and other domains) from the database of the Egyptian licensing authority "Industrial Development Authority (IDA)" and contacted the owners/managers (in-charge persons). While identifying the MSME firms, we did not limit them to a certain level of DEP usage to accommodate the diversity of the firms in terms of their business domain at varying levels of DEP usage. Out of 211 MSME owners/managers, 173 responded to our survey, and the data were then processed to screen for missing, unattended, and outlier responses which resulted in the discarding of nine responses, and the study fetched its final useable sample of 164 firms. The demographic profiles of the firms are exhibited in Table 5.

The scale items of digital entrepreneurial platforms (DEP) and entrepreneurial performance (EP) were pre-emptively analyzed by applying EFA with PCA extraction method having oblique rotation. Results from factor analysis confirmed that the rule of eigenvalues >1 suggested by (Hayton et al., 2004) was observed, thereby suggesting five factors to be retained-four of which relate to DEP and one for EP. A participant-to-item ratio of 5:1 was maintained to see the adequacy of sample size and item communalities were computed. Now it is easy to note that all the communalities are above 0.60. Furthermore, the Kaiser-Meyer-Olkin measure of sampling adequacy came out to be 0.928 which is well above the minimum required of 0.70 by Williams et al. (2010). The results of Bartlett's test of sphericity indicated rejection of the null hypothesis at a chi-square statistic of 3183.562 and at any $p <$ 0.05 level Williams et al. (2010). Factor analysis revealed that 28 items have associations with five distinct factors explaining 64.706 % of the variance; see Table 6. Table 7 shows item loadings, and cross-loadings, for each of the five factors.

Table 5
Respondent's demographic profile.

Variable name	Category	EFA sample (N = 164)		CFA sample (N = 238)	
		Frequency	***%***	***Frequency***	***%***
Company/ business size	Micro	76	46.3	112	47.1
	Small	52	31.7	73	30.7
	Medium	36	22.0	53	22.3
Sector	*Agriculture*	20	12.2	28	11.8
	Food and beverage	18	11.0	28	11.8
	Commerce and retail	35	21.3	49	20.6
	Health and wellness	15	9.1	20	8.4
	Real estate	14	8.5	21	8.8
	Knowledge and ICT	17	10.4	23	9.7
	Education	16	9.8	25	10.5
	Others	29	17.7	44	18.5
Age of business	*Less than 5 years*	54	32.8	82	34.5
	6-10 years	71	43.4	98	41.2
	Above 10 years	39	23.8	58	24.4

Table 6
Total variance explained.

Factor	Eigenvalue	% of variance	Cumulative variance %
1	14.299	46.125	46.125
2	2.506	8.084	54.209
3	1.246	4.018	58.226
4	1.185	3.499	61.726
5	1.024	2.981	64.706

3.4. Stage 4: Scale validation through CFA

For validation and finalization of the scales, we used CB-SEM-based CFA in AMOS 24.0 (Anderson & Gerbing, 1988). Scale validation process included the assessment of convergent validity (model-fit indices, and convergent validity), and divergent validity. For convergence, we considered model-fit indices, item loadings, AVE values, and composite reliability measures. In the process, we also assessed the discriminant validity using two approaches – Fornell and Larcker (1981) and HTMT ratio (Henseler et al., 2015). Since digital entrepreneurial platforms (DEP) is a higher-order reflective construct being reflected by four lower-order constructs, we ran two CFA models – first-order and second-order. For this purpose, we conducted another survey using scale items finalized by EFA at stage 3. Adequate sample size was determined using the a-priori sample size calculator for SEM by Daniel Soper (Soper, 2024; Westland, 2010), with an effect size of 0.30, statistical power at 0.80, and probability level at 5 %. The calculator fetched a recommended minimum sample size of 233 valid responses. In adherence to the required minimum sample size, we again took data from 238 MSME entrepreneurs in Egypt (screened data sample after taking care of missing, unengaged, and outlier responses) from June to September 2024. This time, the data was collected from altogether different entrepreneurs than the ones contacted for EFA (dimensionality reduction at stage 3) to ensure the distinctiveness of the data so that generalizability is established.

From the results of the first-order CFA model, we found the CR values falling in the range of 0.842–0.896 which is above the threshold of 0.70 (Hair et al., 2019). The model-fit indices (viz. GFI, TLI, CFI, and RMSEA) were also found to be well within/above the suggested thresholds (see Table 8). Convergent validity was established by ensuring that AVE values are above 0.50 (see Table 9) (Henseler et al., 2015). In order to ensure the convergence of first-order latent constructs—opportunity identification and validation, resource mobilization, and sharing,

Table 7
Pattern matrix with cross-loadings.

	Factor				
Scale item with code	OIV	EP	FI	RMS	MCE
OIV1: Digital platforms enable me to identify and capitalize on new business opportunities that may not have been accessible otherwise."	***0.639***	−0.070	−0.159	0.157	0.086
OIV2: I rely on digital platforms to validate my initial business ideas."	***0.782***	−0.005	−0.118	0.064	0.135
OIV3: Digital platforms provide tools that are effective for scanning and assessing business opportunities.	***0.637***	−0.063	0.195	0.178	0.151
OIV4: Digital platforms give me the opportunity to expand my business.	***0.852***	−0.046	−0.121	0.074	0.207
OIV5: Digital platforms have opened up entrepreneurial opportunities for me.	***0.740***	0.079	0.081	−0.168	0.108
OIV6: Digital platforms have inspired me to pursue my own entrepreneurial endeavor.	***0.769***	−0.170	0.069	−0.012	0.129
RMS1: Digital platforms enable me to access crucial resources	−0.121	0.013	0.110	***0.636***	0.054
RMS2: Digital platforms provide me technical capabilities needed to develop and refine my business offerings.	−0.189	0.035	0.121	***0.759***	0.134
RMS3: Digital platforms have allowed me to expand my reach to supportive community of experts or entrepreneurs who can provide feedback and advice.	0.041	0.073	0.149	***0.653***	0.083
RMS4: Analytics and performance metrics from these platforms are vital for monitoring and steering my business's success.	−0.135	0.111	−0.129	***0.848***	−0.077
RMS5: Digital platforms provide valuable knowledge and information.	0.153	−0.169	0.016	***0.643***	−0.058
RMS6: Some digital platforms represent powerful source of capital	0.128	−0.064	−0.002	***0.615***	0.020
RMS7: Digital platforms help me collaborate closely and share resources.	0.061	−0.091	0.193	***0.694***	−0.059
RMS8: Digital platforms help me to educate myself and stay updated on the latest trends and advancements in the digital space	0.096	−0.148	0.119	***0.648***	−0.054
FI1: Digital platforms are essential in facilitating the creation of innovative products	−0.204	0.033	***0.846***	0.092	0.008

(*continued on next page*)

Table 7 (*continued*)

Scale item with code	Factor				
	OIV	EP	FI	RMS	MCE
and services in my venture.					
FI2: I turn to digital platforms for innovative solutions when faced with entrepreneurial challenges.	0.162	0.005	***0.654***	−0.066	0.074
FI3: Digital platforms inspire me to establish innovative solutions that operate entirely	0.007	0.090	***0.802***	−0.009	0.106
FI4: Digital platforms have been instrumental in helping me and my business identify challenges and create new innovations.	0.145	0.039	***0.649***	0.177	−0.002
MCE1: Digital platforms play a crucial role in my marketing strategies, enabling me to effectively connect with and engage potential customers.	−0.123	0.094	0.135	0.087	***0.618***
MCE2: The platforms provide the tools and channels necessary for effective product commercialization.	0.094	0.179	0.238	−0.092	***0.739***
MCE3: I'm confident in utilizing digital platforms to expand my customer base.	−0.074	0.047	0.199	0.193	***0.649***
MCE4: I'm confident in utilizing digital platforms to introduce my business ideas to the market.	−0.156	0.058	0.061	−0.075	***0.741***
MCE5: Digital platforms enable me to establish an online presence, showcase my services, and connect with customers effectively.	−0.047	0.130	0.192	−0.145	***0.730***
EP1: Over the three years, my business has witnessed a significant year-on-year growth in profit.	−0.006	***0.726***	−0.037	0.150	0.074
EP2: Over the three years, my business has witnessed a significant year-on-year growth in sales.	0.131	***0.707***	−0.023	0.001	0.116
EP3: Over the three years, my business has achieved a higher market share compared to its peers.	0.020	***0.709***	0.003	−0.036	−0.080
EP4: Over the three years, my business has witnessed a significant increase in the number of products/services being offered.	−0.167	***0.748***	0.112	0.087	−0.096
EP5: Over the three years, my business has witnessed a significant improvement in the cost efficiency of production activities.	−0.125	***0.770***	0.043	−0.038	−0.036

Note: Extraction Method: Principal Axis Factoring. **Rotation Method:** Promax with Kaiser Normalization. **OIV** = Opportunity identification and validation; **EP** = Entrepreneurial performance; **FI** = Fostering innovation; RMS = Resource mobilization and sharing; **MCE** = Marketing and customer engagement.

Table 8
CFA models fit indices.

Model	CMIN/DF	GFI	TLI	CFI	RMSEA
Model-1 (First-order)	1.855	0.931	0.966	0.978	0.046
Model-2 (Second-order)	1.877	0.922	0.963	0.974	0.049
Recommended value	Acceptable 1-4	≥0.90	≥0.95	≥0.95	<0.07
	Wheaton et al. (1977)	Shevlin and Miles (1998)	Hu and Bentler (1999)	Hu and Bentler (1999)	MacCallum et al. (1996)

Table 9
CFA loadings, Cronbach's alpha, CR, and AVE.

Variable name	No. of items	Avg CFA loading	Alpha (α)	CR	AVE
Digital entrepreneurial platforms (second-order)	–	0.823	0.875	0.880	0.677
Opportunity identification and validation	6	*0.757*	*0.842*	*0.848*	*0.573*
Resource mobilization and sharing	8	*0.745*	*0.890*	*0.896*	*0.554*
Fostering innovation	4	*0.793*	*0.865*	*0.872*	*0.629*
Marketing and customer engagement	5	*0.802*	*0.896*	*0.902*	*0.643*
Entrepreneurial performance	5	0.766	0.876	0.881	0.587

Note: Average variance explained (AVE) was calculated to corroborate the convergent validity of each latent construct.

fostering innovation, and marketing and customer engagement with the second-order latent construct digital entrepreneurial platform (DEP), we ran a second-order CFA and the results were found satisfactory affirming the appropriateness of model-fit indices, the sufficiency of AVE and CR values (see Tables 8 and 9).

In the context of scale validation through CFA, there must be enough divergence among the latent constructs. We assessed divergent validity by using two methods. First, we followed the procedure advanced by Fornell and Larcker (1981), according to which a latent construct has adequate divergence if its Average Variance Extracted value is greater than its bivariate correlation with other latent constructs. From Table 10, results affirm ascertain that AVE values (in bold on-diagonal) of each latent construct are considerably higher than the correlation coefficients (off-diagonal values) as mentioned below, hence ensuring divergent validity.

Second, we used the HTMT ratio approach suggested by Henseler et al. (2015) to assess the divergent validity. The assumption of divergent validity is considered to be fulfilled when HTMT ratios among the latent constructs are below the threshold of 0.85 (Henseler et al., 2015). Table 11 affirms that for none of the latent constructs, the HTMT ratios breach the threshold of 0.85; thus, fulfilling the assumption of divergent validity based on HTMT ratio criteria as well.

3.5. Stage 5: Establishing nomological validity and testing the hypotheses

Post-scale validation through CFA, we followed the recommendation of Podsakoff and MacKenzie (1994) and checked for nomological validity of DEP and EP scales by checking the direct linkage between DEP and EP (adhering to one of the study's objectives). Since DEP is a higher-order construct being reflected by four lower-order constructs, we used

Table 10
Correlations, divergent validity, and descriptive statistics.

Construct name	Mean	SD	Skew	OIV	RMS	FI	MCE	EP
OIV	4.432	1.401	1.118	**0.757**				
RMS	5.339	1.339	−1.307	0.632***	**0.802**			
FI	4.884	1.119	−0.892	0.661***	0.670***	**0.793**		
MCE	5.102	1.287	0.990	0.587***	0.626***	0.691***	**0.745**	
EP	5.419	1.303	1.571	0.520***	0.515***	0.682***	0.648***	**0.766**

Note: **Correlations are significant at 1 % level (i.e., p-value <0.01. Values in bold on diagonals represent the square root of AVE, and off-diagonal values represent correlations. **OIV** = Opportunity identification and validation; **RMS** = Resource mobilization and sharing; **FI** = Fostering innovation; **MCE** = Marketing and customer engagement; **EP** = Entrepreneurial performance.

Table 11
Discriminate validity (HTMT ratio criterion).

Construct name	OIV	RMS	FI	MCE	EP
OIV					
RMS	0.593				
FI	0.631	0.705			
MCE	0.672	0.693	0.764		
EP	0.662	0.608	0.554	0.734	

OIV = Opportunity identification and validation; **RMS** = Resource mobilization and sharing; **FI** = Fostering innovation; **MCE** = Marketing and customer engagement; **EP** = Entrepreneurial performance.

DEP as the second-order construct while checking its effect on EP. The results reveal that DEP has a positive and significant influence on EP (β = 0.689; S.E. = 0.113; C.R. = 6.896; CIs at 95 % = 0.520, 0.804) with an adjusted R^2 value of 0.475 asserting an explanatory power of 47.50 %. These results confirm that the tested linkage aligns with the hypothesized direction; therefore, the proposed hypothesis (H1) is accepted, and the developed and validated scales demonstrate nomological validity.

4. Conclusion and implications

This research had a twofold purpose. The first aim was to develop and validate a multidimensional scale to measure the capability of digital entrepreneurial platforms. This plays an essential role in filling a significant gap in literature by providing a scale to evaluate how entrepreneurs use these platforms to facilitate key processes and activities within the digital entrepreneurship ecosystem. Secondly, this research also sought to explore how the capabilities of digital entrepreneurial platforms in enhancing entrepreneurial performance, particularly in areas such as profitability, increased market share, and growth of product or service portfolio. The current study contributes to the literature on digital entrepreneurship in two important areas. First, this paper comes to fill a research gap that there has been a need to conduct a development and validation research of digital entrepreneurial platform capability instrument based on entrepreneurial processes and activities. Second, the capabilities of digital entrepreneurial platforms (DEP) that have been conceptualized and validated in this study advance the understanding of how these platforms facilitate entrepreneurial activities while being operationalized as a higher-order construct integrating the latent concepts viz. opportunity identification, resource mobilization, innovation, and customer management into a unified framework. Third, through the development and validation of multidimensional DEP capability scales, the study provides the stakeholders with a robust and empirically grounded unified framework to diagnose and identify the pivotal digital platforms for enhanced entrepreneurial performance.

Through the development and validation of DEP capability scales, the study also offers several practical implications. The MSME owners and/or managers can utilize the DEP capabilities scales to identify the strengths and weaknesses across the key dimensions viz. opportunity identification, resource mobilization, and fostering innovation and marketing, and map their developmental strategies based on data-driven insights. Policymakers for MSMEs can leverage DEP capabilities scales to devise evidence-driven policy interventions to develop tailor-made digital literacy programs. These DEP capability scales can also be of benefit to the digital platform developers in strategizing their digital platform offerings through the embedment of the features that align with DEP capability platforms pivotal for MSME's performance. By adopting the validated DEP capability scales for decision-making, the stakeholders can move to a capability-driven structured approach.

4.1. Limitations and future research

Although we ensured the utmost care while carrying out the study, it still suffers from several limitations. First, the context of the study is limited to Egyptian MSMEs; hence, limiting the generalizability of the scales. In the future, studies may be conducted by taking a sample from other contexts/countries to ensure geographical and cultural heterogeneity. Second, since the study has validated the DEP capability scales by taking the self-reported data from MSME owners/managers, it could be affected by social desirability biases; thus, future studies may be conducted for scale refinement by integrating digital platform usage analytics as objective metrics for complementing the survey responses. The use of cross-sectional design to establish the linkage between DEP capabilities and entrepreneurial performance (EP) obscures causal inferences; therefore, a longitudinal study design could be adopted by future studies to establish the link between DEP capabilities and EP. Lastly, the exclusion of external environmental factors (viz. regulatory environment, quality of digital infrastructure) leaves room for future studies to incorporate such variables.

CRediT authorship contribution statement

Khalid Mady: Writing – review & editing, Writing – original draft, Methodology, Formal analysis, Conceptualization. **Imran Anwar:** Writing – review & editing, Writing – original draft, Validation, Methodology, Formal analysis. **Mohamed Aboelmaged:** Supervision, Project administration, Investigation. **Mohamed Ahmed Sulub:** Visualization, Software, Resources, Data curation.

Appendix

Appendix A
Final validated measurement scales with dimensions and items

Opportunity identification & validation
1. Digital platforms enable me to identify and capitalize on new business opportunities that may not have been accessible otherwise.
2. I rely on digital platforms to validate my initial business ideas.
3. Digital platforms provide tools that are effective for scanning and assessing business opportunities.
4. Digital platforms give me the opportunity to expand my business.
5. Digital platforms have opened up entrepreneurial opportunities for me.
6. Digital platforms have inspired me to pursue my own entrepreneurial endeavor.
Resource mobilization & sharing
1. Digital platforms enable me to access crucial resources
2. Digital platforms provide me technical capabilities needed to develop and refine my business offerings.
3. Digital platforms have allowed me to expand my reach to supportive community of experts or entrepreneurs who can provide feedback and advice.
4. Analytics and performance metrics from these platforms are vital for monitoring and steering my business's success.
5. Digital platforms provide valuable knowledge and information.
6. Some digital platforms represent powerful sources of capital.
7. Digital platforms help me collaborate closely and share resources.
8. Digital platforms help me to educate myself and stay updated on the latest trends and advancements in the digital space
Fostering innovation
1. Digital platforms are essential in facilitating the creation of innovative products and services in my venture.
2. I turn to digital platforms for innovative solutions when faced with entrepreneurial challenges.
3. Digital platforms inspire me to establish innovative solutions that operate entirely
4. Digital platforms have been instrumental in helping me and my business identify challenges and create new innovations.
Marketing & customer engagement
1. Digital platforms play a crucial role in my marketing strategies, enabling me to effectively connect with and engage potential customers.
2. The platforms provide the tools and channels necessary for effective product commercialization.
3. I'm confident in utilizing digital platforms to expand my customer base.
4. I'm confident in utilizing digital platforms to introduce my business ideas to the market.
5. Digital platforms enable me to establish an online presence, showcase my services, and connect with customers effectively.

Data availability

The research data is part of an ongoing project and is subject to further analysis and investigation.

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